

However, only the lower midfacial area can be advanced, and it is impossible to resolve the overall midface deficiency due to the position of the Le Fort I osteotomy line.^{5,9} Thus, facial implant placement can be considered because the upper-midface deficiency cannot be resolved through Le Fort I osteotomy.

Various types of prosthesis, including silicone (polydimethylsiloxane), GoreTex (expanded polytetrafluoroethylene, WL Gore & Associates Inc, Flagstaff, AZ), and MedPor (high-density porous polyethylene, Porex Industries, Fairburn, GA), have been used to alleviate facial depression. Since these prostheses are supplied by sizes, premanufactured prostheses of an appropriate size should be selected during the surgery and need to be carved before placing them. Consequentially, the prostheses are not passively fitted on the bone surface, and the gap is formed, leading to a risk of prosthesis migration to an unwanted area. Also, since prostheses are often placed based on the surgeon's tactile sense and experiences, there are possibilities of malpositioning or under-/over-correction.¹⁰

However, the prosthesis, manufactured using CAD/CAM and 3D printing technology can be passively seated on the patient's bone surface without gap. By forming screw fixation holes in advance during the manufacturing procedure, the position of the prosthesis can be confirmed during the fixation process, and the migration of the prosthesis can be prevented by stably fixing it. Furthermore, the surgeon can decide the thickness of the prosthesis based on the patient's facial contour, and the possibility of under-/overcorrection can be significantly minimized. Finally, since the fixation holes of the prosthesis are designed and set to avoid major anatomical structures such as nerve trunk, there is little possibility of complications such as nerve damage.¹¹

Here, implants made of polyetherketoneketone were used to relieve the midface deficiency of a patient with Crouzon syndrome. First, through virtual surgery, the postoperative position of the patient's maxilla and mandible was checked, and then the implants covering the patient's midface were designed accordingly. After designing customized plates for maxillary fixation, plate-supported surgical stents were designed to indicate the holes where the implant fixation screws would be located. Before placing implants using the prepared surgical stents, screw holes were formed in the maxillary anterior wall, and the location of the implants was double-checked by placing them in accordance with these holes.

To shorten the operation time and improve the accuracy and predictability of surgeries, the development of 3D virtual surgery software, CAD/CAM and 3D printing technology has been applied in various fields of surgery. In the field of maxillofacial surgery, 3D technology has been applied to maxillofacial reconstruction and orthognathic surgeries to obtain more esthetic surgical outcomes. Even with this case presented, if the surgery were performed by the conventional method, the operation might not have been done as planned. Particularly, the application of new technology can be advantageous to the operator when it comes to the prediction of hard tissue movement through virtual simulation during patient diagnosis or treatment planning, not to mention the utilization of 3D virtual surgery program and surgical materials fabricated using CAD/CAM and 3D printing technology.

In this case report, patient-specific facial implants using 3D virtual surgery, CAD/CAM and 3D printing technology were reviewed. As shown in the postoperative cone-beam computed tomography data and clinical photos, the application of these latest technologies certainly increases the surgical accuracy and guarantees better results. Although long-term stability studies have not been conducted, the latest technology applied to surgery is expected to be better than the 1 with the conventional method. Thus, it is necessary to strive to pursue better surgical outcomes using various cutting-edge technologies in the field of oral and maxillofacial surgery.

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Long-Term Stability of Alveolar Bone Graft in Cleft Lip and Palate Patients: Systematic Review and Meta-Analysis

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Objective: Alveolar bone grafts are the golden standard in treating patients with oral cleft and hence, the long-term success of this

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treatment is the subject of many studies. The aim of this study was to systematically review literature that examined the long-term stability of alveolar bone grafting in patients with cleft lip and palate.

Methods: In this study, data was collected via a comprehensive search of random and nonrandom studies evaluating the success of bone grafts in patients with cleft lip and palate in ISI Web of Science, EMBASE, Scopus, Medline, and Cochrane Central Register of Controlled Trials from the beginning until July 3, 2020. Also, a search was conducted for ongoing studies in the trial registry and a hand search of related gray literature and references of the relevant studies. The relevant data of the included studies were extracted using predefined extraction forms. Meta-analysis was performed using inverse-variance for calculating the pooled outcome of outcome measures. The pooled mean success rate of alveolar bone graft and mean difference of cleft volume after 1 year was presented as the outcome of the analysis.

Results: Eight studies (1 randomized clinical trial study and 7 controlled prospective studies) were included in this meta-analysis. The results of the analyses showed that the total percentage of secondary alveolar success in the follow-up period of at least 1 year according to the Bergland index was 76.52% (pooled success = 76.52%, 44.92–98.10). Also, bone defect in the cleft area reduced significantly after bone grafting (MD = 0.62 cm³, 0.48–0.75, $P < 0.00001$). The total percentage of bone filling after 1 year and according to CBCT (cone beam computed tomography) was about 63.38%, and the rate of definite natural eruption of canine within the graft was 80.89%.

Conclusions: The total success rate of secondary alveolar bone grafting in the follow-up period of at least 1 year was clinically significant and alveolar bone grafting significantly reduced bone defect volume. Also, there was no difference in the amount of bone remaining after 1 year in patients whose canine had erupted through the grafted bone and in those whose canine had not erupted.

Key Words: Alveolar bone graft, Cleft lip and palate, Long-term stability, Success rate

Oral clefts are one of the most common deformities of the craniofacial region.¹ The prevalence of cleft lip and palate varies in different populations and is reported between 1 and 1.5 per 1000.^{2,3} Various hereditary and environmental factors play a role in the occurrence of facial clefts in a way that their etiology is considered as multifactorial. Hereditary factors, such as race and ethnicity and some genes, and environmental factors, such as maternal hormonal disorders,⁴ use of neuropsychiatric drugs,⁵ vitamin deficiency⁶ and folic acid, hypoxia, and smoking,⁷ are considered as etiological factors.

The management of cleft lip and palate patients requires a multidisciplinary approach that includes surgical and nonsurgical treatments.^{8,9} The goal of surgical interventions in the treatment of cleft lip and palate is to restore morphology and function without impairing growth potential. Secondary alveolar bone grafting is one of the main challenges during orthodontic treatment and surgery of patients with cleft lip and palate.¹ The benefits of alveolar bone grafting include integrating maxillary segments and helping to create a uniform alveolar ridge, supporting and assisting bone growth in adjacent teeth, an opportunity for rapid maxillary suture expansion, permitting orthodontic tooth alignment, allowing placement of osseointegrated

implants, removal of oronasal fistula, and improving alar base symmetry, lip support, and facial esthetics.^{1–3,9–12}

Alveolar bone grafting can be performed at various ages based on the patient's age or dental development. There are 2 categories in this regard: primary bone graft and secondary bone graft. In secondary bone grafting, therapeutic and surgical procedures are performed after cleft palate repair, whereas in primary bone grafting, these stages are performed before or during palatoplasty.^{13,14} Long-term evaluations of primary alveolar grafting indicate serious abnormalities in maxillary growth, such as midface deficiency and anterior crossbite. Secondary alveolar grafting is divided into 3 categories: 1) Early, which is done usually between 2 and 5–6 years old, before the permanent teeth erupt and after the completion of the deciduous dental system; 2) conventional graft which is performed in the mixed dentition, and 3) late graft which is done after the completion of the permanent dentition.¹⁵ It is very useful to do alveolar bone grafting for early mixed dentition at ages 6 to 8^{16–18} because the maxillary transverse growth is almost complete and the first molars are in the occlusion. The patient cooperation in orthodontic interventions, the sufficient bone density and minimal morbidity in the donor site, fast recovery after surgery and the grafted bone allowing the incisors to erupt, are other beneficial factors for alveolar bone grafting at this age.^{18,19} However, there is sufficient evidence that the following alveolar bone grafting, tertiary grafting, may be required.^{20,21}

Source of Bone Graft Materials and Their Substitutes

Both cortical and cancellous bones can be used for reconstruction of alveolar bone clefts, but most surgeons prefer cancellous bone because of the cell transfer and revascularization in osteoinduction and osteoconduction. Different bone graft materials including autologous, allogeneic, and xenogeneic bone materials, recombinant human bone morphogenetic proteins, and growth factors have been used for reconstruction of alveolar cleft. Of these, fresh autologous cancellous bone is the ideal bone graft source.¹⁵ Recently, some clinicians have been used autologous bone graft substitutes such as bone morphogenetic proteins (BMPs) for the reconstruction of the alveolar cleft. Bone morphogenetic proteins are a group of osteoinductive polypeptides that stimulate new bone formation by differentiation of stem cells to osteoblasts.^{22,23}

Alveolar Bone Graft Surgery

Based on the routine surgical procedure, under general anesthesia and vasoconstrictor injections, gingival mucoperiosteal flaps are designed around the margins of the cleft and elevated medial and lateral mucoperiosteal flaps are generated from the cleft and the gingival sulcus of the teeth. For sufficient movement, the flap must be extended to the first or second molar and to the buccal sulcus. These flaps rise around the piriform and detach from the nasal mucosa. The palatal flaps also separate from the palate around the margin of the cleft. After adequate exposure, the nasal lining of the nostril floor is approximated and sutured, and the palatal flaps are sutured together to form a soft tissue pocket. After placing the blocks and bone particles and compacting them in this pocket, the mucoperiosteal gingival flaps are transposed without stretching to cover the area. In the case of bilateral clefts, the orthodontic alignment of the premaxilla is set and the cleft is repaired using a method similar¹⁵ Fig. 1.

Evaluation of Alveolar Bone Graft Results

Radiographic evaluation helps to determine the outcome of the bone grafting by assessing the bone fill of the defect, eruption status

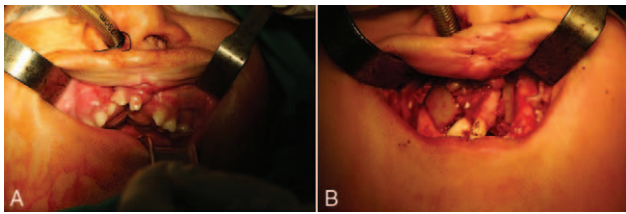


FIGURE 1. (A) A bilateral alveolar cleft. (B) Alveolar bone graft using cancellous and cortical bones.

of the lateral incisor or canine adjacent to the cleft site, and adequacy of bone volume for placement of endosseous implants. Generally, two-dimensional (2D) radiographs including panoramic, occlusal, and periapical films are used to assess pre- and post-surgical conditions to determine the success of alveolar bone grafting. There are several scoring systems that evaluate the success of the alveolar bone grafting by measuring the height of the remaining interdental bone based on 2D radiographs.^{11,24,25} The Bergland scoring system¹¹ has remained the gold standard for 2D assessments of bone graft success. According to this criteria, Bergland score one indicates normal height and score two indicates 3 quarters of normal height, score three indicates 1 quarter of normal height, and score four indicates no bone and graft failure in the cleft area.

Conventional computed tomography scans have also been used to assess outcome of alveolar bone grafting.^{26–30} Lee et al²⁶ found that with dental radiographs, the number of clinically successful bone grafts were overestimated by 17% as compared with conventional computed tomography scans. This inaccuracy was attributed to the fact that 2D radiographs cannot show depth and volume of residual bone grafted.³¹

Many studies have been performed to evaluate the success rate of alveolar bone grafting in patients with cleft lip and palate based on clinical or radiographical scales, reporting mixed results.^{12,32–38} However, only few studies have investigated the long-term stability of alveolar bone grafts.^{34,39} The aim of this study was to systematically review the literature that examined the long-term stability of alveolar bone grafting in patients with cleft lip and palate.

METHODS

Data Sources and Search Strategy

A systematic review and meta-analysis study was performed in accordance with the recommendations of the Cochrane Handbook for Systematic Reviews of Interventions (version 5.1.0) and the MOOSE guidelines for Meta-Analysis and Systematic Reviews of Observational Studies. In this study, data were collected from studies that evaluated the success of alveolar bone graft in patients with cleft lip and palate.

All random and nonrandom articles related to the topic and without language restrictions were searched in the databases of ISI Web of Science, EMBASE, Scopus, Medline, and Cochrane Central Register of Controlled Trials using MeSH headings and related words and search strategy from the beginning to July 3, 2020. A summary of the search strategy is shown in Supplementary Digital Content, Table 1, <http://links.lww.com/SCS/D393>. Furthermore, the abstracts and proceedings of IADR congresses, WHO Clinical Studies Registry, and clinicaltrials.gov were reviewed for ongoing studies. The references of relevant reviews, meta-analysis, and references, and the list of all of the included studies were also manually searched. If a study in a language other than English was found, that study was first translated into English and then reviewed.

Inclusion and Exclusion Criteria

According to the PICO process, the **population** under study was patients with unilateral or bilateral cleft lip and palate. **Intervention** was defined as secondary alveolar bone grafting, **comparison** group were patients before or without graft and other methods were performed for grafting, and **outcome** was defined as the long-term success of the alveolar bone graft.

Inclusion criteria: Clinical human studies, prospective studies, evaluation of graft success rate for at least 1 year, and radiographic evaluation of graft success rate were considered.

Exclusion criteria: Studies on patients with syndromic cleft lip and palate, evaluation of success on recurrent fistulas, and concomitant alveolar bone grafting with maxillary osteotomy were excluded from the present study.

Study Selection

All the obtained articles were entered into the software Endnote (version 20, Clarivate Analytics, Philadelphia, US), and after removing the duplicate studies, the titles and abstracts of the remaining studies were reviewed by 3 authors (E.B, A.J, and E.K). All abstracts were reviewed, and each relevant article was marked for further evaluation. The full text of the relevant studies was retrieved and the studies that met the present study's inclusion criteria were included in the analysis. The references of each eligible article were also screened for relevant articles. The search was performed independently by 2 authors.

Study Quality Assessment

The quality of studies eligible to enter the meta-analysis was assessed by the Newcastle-Ottawa evaluation systems for nonrandomized studies and Cochrane Risk of Bias (RoB) for randomized clinical trial (RCT) studies by 2 of the authors (A.J and E.K). Using the Newcastle-Ottawa scoring system, each study was scored based on 3 general items: 1) study group selection bias, 2) confounding variables bias, and 3) measurement bias. Selection bias was further divided into subgroups of sample size selection and comparison group selection, ensuring that the comparison group was affected by the intervention and the result was not clear from the beginning of the study. Measurement bias was also divided into evaluation of outcome, follow-up length, and percentage of patients present in the follow-up. The maximum score a study could receive was 9. Scores less than 6 indicated high risk studies, 6–8 indicated medium risk studies, and 9 indicated low risk studies. In the Cochrane RoB randomization, allocation concealment, blinding of participants and evaluators, incomplete results, and selected data report bias were divided into low risk, high risk, and unknown risk according to their RoB. If there was a high or unknown risk for each of the indicators, the RoB in the whole study was determined as high or unknown, respectively. In case of disagreement, it was resolved through negotiations between the 2 authors.

Data Extraction

The relevant data from each study were extracted using pre-defined forms. Using these forms, the desired data extracted from all studies included: year of publication, country, type of study, number of subjects in the case and control groups, sex of subjects in the study, average age of subjects in the study, type of cleft, duration of the study, time of alveolar bone grafting, preoperative interventions (primary alveolar bone grafting, gingivoperiosteoplasty, and nasolabial molding), cleft width, eruption stage of tooth adjacent to the cleft, orthodontic treatment (fixed orthodontics and maxillary

expansion) before or after grafting, radiography used to evaluate success rate, bone volume scales performed, graft success rate, and number of patients related to each of the bone volume scales.

In this study, several analyses were performed including the volume of residual bone defect after the alveolar bone graft, the percentage of bone filling of the defect area, the percentage of canine growth in the alveolar bone graft, and the effect of canine growth on the bone filling rate of the defect.

If only the mean bone volume remaining after graft was reported in baseline studies, the mean residual defect was calculated by the following formula and then used in meta-analysis:

$$MD = M1 - M2$$

$$\sigma D = \sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{n}}$$

In this formula, MD is the mean residual defect, M1 is the preoperative defect, and M2 is the residual bone volume. Also, σD is the standard deviation of the residual defect, σ_1 is the standard deviation of the defect before surgery, σ_2 is the standard deviation of the remaining bone volume, and n is the number of patients in the study group.

Statistical Analysis

The success rate of alveolar graft using the Bergland Scale, the volume of remaining defect after 1 year, and the percentage of remaining graft compared to the initial volume was used as the outcome measures of the analyses. Pooled mean success rate and pooled mean ratio for the remaining graft was used in calculating the result of the studies using these measures and mean difference (MD) was used to compare the volume of defect before graft surgery and after 1 year.

Meta-analysis was performed using inverse-variance random-effects analysis and the software Review Manager (version 5.4, Copenhagen, Denmark, Cochrane Collaboration, 2019). Chi squared q test and Tau² were used to evaluate the presence or absence of heterogeneity between studies and the amount of this heterogeneity was determined by I^2 .

Investigation of Publication Bias

Due to the presence of less than 10 studies in the analysis of each variable, publication bias was not examined by funnel plot symmetry.

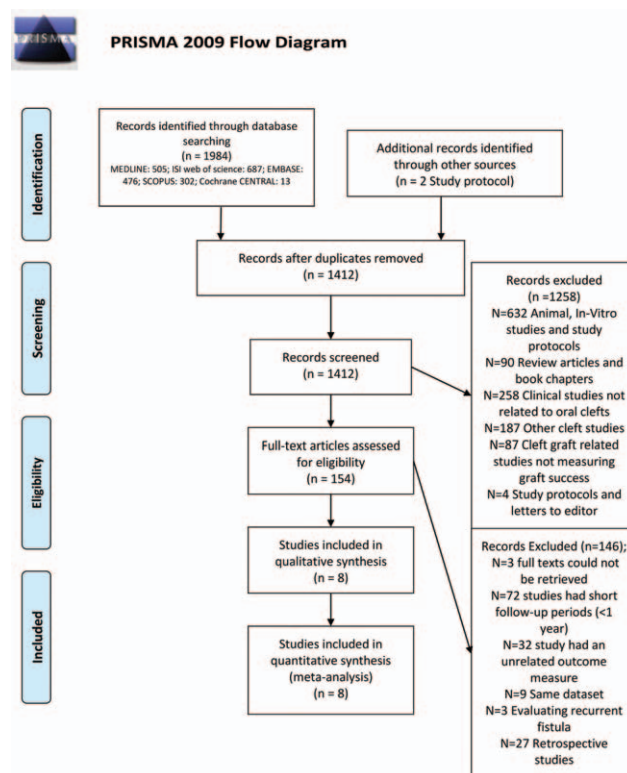
Sensitivity Analyses

Sensitivity analysis was performed to investigate the effect of confounding variables by deleting the study with that variable and performing the meta-analysis again and comparing the results of the 2 meta-analyses.

RESULTS

Selection of Studies

A total of 1984 studies related to the subject were found by searching databases. After eliminating duplicate studies, 1412 studies and an additional 3 studies that were found by hand search were evaluated using their titles and abstracts. After reviewing the titles and abstracts, 1258 studies including animal and in vitro studies (632), review studies and book chapters (90), clinical studies unrelated to cleft palate (258), cleft palate studies with variables other than alveolar bone graft success rate (87), letters to the editor, and study protocol (4) were excluded. The full texts of the



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FIGURE 2. The selection process of studies that achieved long-term stability and examined the alveolar bone in patients with cleft lip and palate.

remaining 154 studies were reviewed. The following were removed from the list of articles: 72 studies that evaluated the success of the graft for less than 1 year, 32 studies with unrelated variables, 3 studies that included patients with recurrent fistulas, 9 studies that used same datasets as newer studies, and 3 studies due to the inability to receive the full-texts. After removal of 27 retrospective studies, 8 articles (1 RCT) and 7 controlled clinical trials were included in the study. These articles were published between 2000 and 2017 (Fig. 2).

Study Characteristics

Supplementary Digital Content, Table 2, <http://links.lww.com/SCS/D393> shows the information extracted from the articles. In these studies, a total of 199 patients and 231 cleft lip and palate (30 bilateral clefts and 171 unilateral clefts) were included. Average age of patients was 9.87 years old, and the follow-up period of the studies was 1 to 3 years. In all studies, the alveolar bone graft donor site was the iliac crest, 6 studies used CBCT to evaluate bone volume before and after grafting, and 1 study used a combination of conventional and CBCT radiographic methods.

Quality of Studies

According to the quality assessment for Newcastle-Ottawa nonrandomized studies, 1 study had a high RoB and 6 studies had moderate RoB. Additionally, according to Cochrane RoB scale, the randomized study had an unclear RoB. Since less than 10 studies

were included in the analysis of each variable, the publication bias analysis was not examined. Supplementary Digital Content, Table 3, <http://links.lww.com/SCS/D393>.

RESULTS

In this study, some analyses were done based on the volume of bone defect remaining after the secondary alveolar bone graft, the percentage of bone filling of the defect, the percentage of canine eruption toward the alveolar bone graft, and the effect of canine eruption on the bone filling rate.

The results of meta-analysis on 5^{29,32,33,40,41} studies and 71 patients showed that the volume of bone defect in the cleft area reduced significantly after bone grafting (MD = 0.62 cm³, 0.48–0.75, $P < 0.00001$) (Fig. 3A). In this analysis, the heterogeneity was moderate between studies ($I^2 = 55\%$) after performing sensitivity analysis and omitting the Tai et al's²⁹ study due to the presence of 3 patients with bilateral cleft and no heterogeneity that remained between studies ($I^2 = 0\%$). Hence, the result of the analysis did not change significantly (MD = 0.58 cm³) and bone grafting still significantly reduced the volume of bone defects (MD = 0.58 cm³, 0.49–0.66, $P < 0.00001$) (Fig. 3B).

According to the results of 5 studies^{29,32,33,40,41} on 71 patients, the total percentage of filling the defect area by graft after 1 year is about 63.38% (pooled bone fill percentage = 63.38%, 46.53–80.23) (Fig. 3C).

Similarly, regarding the normal eruption rate of canine teeth, the results of 4 studies^{29,32,41,42} including 105 patients were evaluated and in total, the rate of definite natural eruption of canine was 80.89% (pooled bone fill percentage = 80.89%, 73.04–87.69) (Fig. 3D).

Finally, to evaluate the effect of canine tooth eruption on residual bone graft, a meta-analysis of the results of 2 studies^{29,32} involving 36 patients showed no difference in residual bone between patients with or without canine eruption (MD = −0.85%, 0.85%, −88.01–86.32, $P = 0.98$), although there was a very high degree of heterogeneity between the 2 studies ($I^2 = 98\%$) (Fig. 3E).

DISCUSSION

The Main Finding

Bone grafting significantly reduces the volume of bone defects (MD = 0.58 cm³, 0.49–0.66, $P < 0.00001$).

According to the results of 5 studies on 71 patients, the total percentage of bone filling after 1 year is about 63.38% (pooled bone fill percentage = 63.38%, 46.53–80.23).

The total success rate of secondary alveolar bone grafting according to the Bergland index (residual bone height greater than three-fourth normal height)¹¹ is about 76.52% (pooled success = 76.52%, 44.92–98.10).

Regarding the normal eruption of the canine toward the graft, the definite natural eruption rate of the canine was 80.89% (pooled bone fill percentage = 80.89%, 73.04–87.69).

Finally, there was no difference in residual post grafting between those with canine eruptions and those without (MD = −0.85%, −88.01–86.32, $P = 0.98$).

Comparison with Similar Studies

In the present study, to evaluate the long-term stability of alveolar bone graft in patients with cleft lip and palate, studies that had at least 1 year follow-up duration were selected. This duration is more reliable to assess the stability of the alveolar bone graft. Its success is because the highest rate of bone loss occurs within the first year after bone grafting, and so any evaluation done earlier than 1 year is not optimal.^{29,41}

In our study, the total success rate of secondary alveolar bone grafting according to the Bergland index¹¹ is about 76.52%. The study by Schultze-Mosgau et al⁴² reported a 86% success rate and for the Kawakami et al's⁴³ study, the success rate of bone graft was reported to be 59% 1 year after surgery. The Kawakami et al's study showed a lower rate of bone graft success than similar studies like the study by Sharma et al,³⁷ which evaluated the secondary alveolar bone grafting radiographically and clinically. In this study, 10 patients who underwent secondary alveolar bone grafting for different types of cleft palate treated with autologous cranial graft were evaluated. Post-operative radiographic evaluation showed a 87.5% success rate. However, the small sample size was evident when compared to our study. Tomar et al³⁸ performed autogenous iliac crest bone grafting in all cases, with a success rate of 85% according to the Bergland criterion; however, the study had 6 months follow-up and when compared to the study by Kawakami et al⁴³ and Schultze-Mosgau et al,⁴² the follow-up duration was shorter and the risk of bone resorption was at its highest for up to 1 year.²⁹

The effect of canine tooth eruption on residual bone grafting was also evaluated in the present study, and meta-analysis of the results of 2 studies^{29,32} involving 36 patients showed no differences between the remaining bone fill percentages in patients with or without canine eruption (MD = −0.85%, −88.01–86.32, $P = 0.98$). A very high degree of heterogeneity between the 2 studies ($I^2 = 98\%$) also existed. A possible explanation for the differences in the results of studies is that in the Tai et al's study the volume of the canine tooth itself was not included in the calculation of the residual bone graft, whereas in the Feichtinger et al's study, the



FIGURE 3. (A) Mean difference graph of bone defect volume was performed 1 year later. (B). Sensitivity analysis of bone defect volume after removal of the Tai et al's study. (C). Graph of the total percentage of filling of the defective area by the bone after 1 year. (D). Graph of normal eruption rate of canine teeth to the graft after 1 year. (E). Mean difference diagram of the amount of residual bone grafting after 1 year between patients with and without normal canine eruption.

volume of the canine tooth in the residual bone graft volume was calculated. However, even if the canine tooth volume was taken into account in the Tai et al's study (average 481 cubic mm³ or 53% of the residual defect volume), the rate of resorption would still be higher in the canine eruption group (27.4% vs. 7.8%). In addition, in the Schultze-Mosgau et al's study,⁴² the rate of bone resorption in patients who underwent orthodontic closing of the remaining space of the graft was significantly lower than the rate of bone resorption in those who had the remaining space closed with a prosthesis. Other evidence also suggests that tooth movement is osteogenic.⁴¹

Other Factors Affecting the Result

In almost all of the analyses and studies in this review, only individuals with unilateral clefts were examined, except in the Tai et al's study, in which the defect volume and the percentage of defect filling of the left cleft of 3 patients with bilateral cleft was also examined. Some studies have found that the rate of bone graft resorption in patients with bilateral clefts is higher than in patients with unilateral clefts, and as observed, the amount of defect volume after treatment in the Tai et al's study was higher than similar studies in the meta-analysis. However, after performing a sensitivity analysis, no significant difference was found in the results of meta-analysis with or without this study (0.62 and 0.58 cm³, respectively). Also, in the Tai et al's study, the difference between the volume of the residual defect and the percentage of bone filling of the unilateral and bilateral clefts was examined using the student *t* test and no significant difference was observed between the results of these 2 groups (*P* > 0.8).

In our review, all subjects in the included analyses received a secondary bone graft after canine eruption. In the study of Schultze-Mosgau et al,⁴² the rate of long-term bone graft resorption was not significantly different if the graft was performed before or after canine eruption, which is different from the result of retrospective studies by Rawashdeh and Al Nimri,⁴⁴ Hogan et al,⁴⁵ and Jeyaraj et al.⁴⁶ Also, in the study of Oberoi et al,⁴¹ the effect of canine root development before eruption on graft resorption was investigated and no difference existed in individuals with canine root development of more than 50% and less than 50%. It is noteworthy that some researchers such as Bergland et al,⁴⁷ suggested that it was better to do bone grafting when the development of the canine root was between half and three-fourths, because of the improvement of canine eruption and the elimination of canine eruption abnormalities in addition to helping to maintain bone graft.

This study also evaluated the long-term outcomes of iliac crest bone grafting, which is regarded as gold standard bone grafting in facial clefts because of the availability of adequate volume of cortical and cancellous bones.^{10,15} Nonetheless, the use of this bone is associated with its own morbidities and is known to increase the risks and time of surgery and, therefore, other methods have been introduced to replace bone grafts, especially iliac bone grafts. In the study of Du et al,⁴⁰ the rate of bone resorption 1 year after surgery was compared with that of bone marrow cells and beta-tricalcium phosphate granules. No significant difference existed in the Chelsea Score⁴⁸ and defect volume between the 2 groups, whereas the results were slightly in favor of bone grafts. Also, in a study by Canan et al,³³ the rate of long-term resorption was compared between bone graft and rhBMP-2 groups, which used a collagen sponge impregnated with bone morphogenesis protein 2 for grafting. In this study, there was no significant difference in terms of the residual defect and the percentage of filling 1 year after surgery between the 2 groups, whereas in this study, the defect

volume in the bone graft group was slightly better than the other group. Jahanbin et al²² found low-dose rhBMP-2 combined with autogenous bone graft can be promising to reach predictable results after alveolar reconstruction in cleft lip and palate patients.

In this meta-analysis, the volume of bone defects before and after bone graft surgery in 5 studies was calculated in cm³ using CBCT. Based on the evidence, the accuracy of three-dimensional radiographic evaluations was higher than the conventional radiographic methods, and defect volume examination gave more reliable information about the amount of resorption of the graft.⁴⁹ The use of three-dimensional radiography also allowed the surgeon to accurately assess the amount of graft material needed on the site of the clefts. In addition, assessments of the remaining bone volume allowed for a more accurate assessment of the implant placement. Nevertheless, the gold standard for radiographic evaluations of bone grafts in cleft lip and palate patients is still 2D and traditional methods.^{32,41,50}

Suggestions

Due to the complexity of the treatment of patients with cleft palate and the key role of alveolar bone grafting in the treatment process of these patients, it is necessary to conduct higher quality studies and more RCTs in relation to its long-term stability. Subsequent review studies can also compare newer bone grafting methods and materials with conventional secondary alveolar bone grafting using the iliac crest bone.

CONCLUSIONS

- The total success rate of secondary alveolar bone grafting in the follow-up duration of at least 1 year according to Bergland index is about 76.52% and bone grafting significantly reduces the volume of bone defects.
- The total percentage of bone filling after 1 year and according to CBCT is about 63.38%.
- The normal eruption of canine teeth was taken toward the graft area and in total, the rate of definite natural eruption of canine was 80.89%.

Finally, to evaluate the effect of canine eruption on residual bone graft, no difference existed after 1 year in patients whose canine had erupted and in those without an eruption.

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